

BFSI Domain

Knowledge Guide for Data Science Freshers

Banking | Financial Services | Insurance

About This Guide

This guide is designed to help new data science team members quickly understand the BFSI (Banking, Financial Services & Insurance) domain. It covers key concepts, terminology, regulatory frameworks, and data science applications you will encounter when working with BFSI clients.

How to use: Read through each section in order. Pay special attention to the terminology tables, as clients will use these terms frequently in meetings and documents.

INTERNAL TRAINING DOCUMENT

Section 1: What is BFSI?

BFSI stands for Banking, Financial Services, and Insurance. It is one of the largest and most data-intensive sectors in the world. As a data scientist, you will work with enormous volumes of structured and unstructured data to help these organizations make better decisions, comply with regulations, and serve their customers.

1.1 The Three Pillars of BFSI

Pillar	Description & Examples
Banking	Institutions that accept deposits and give loans. Examples: Retail banks (SBI, HDFC), Central banks (RBI, US Federal Reserve), Investment banks (Goldman Sachs), Cooperative banks.
Financial Services	Broader category including capital markets, asset management, payment processing, brokerage. Examples: Mutual funds (AMFI), Stock exchanges (NSE, BSE), NBFCs (Non-Banking Financial Companies).
Insurance	Companies that provide financial protection against risk. Examples: Life insurance (LIC), General insurance (vehicle, health, property), Reinsurance companies.

1.2 Why BFSI is Unique for Data Science Company

- Extremely high data volumes — millions of transactions daily
- Strict regulatory requirements — data must be audit-ready
- High stakes — errors can cost millions or breach compliance
- Real-time processing — fraud detection must act in milliseconds
- Complex, interconnected entities — customers, accounts, products, channels

Key Takeaway

Every data science project in BFSI must balance performance, interpretability, and regulatory compliance. A model that works well but can't be explained to a regulator may not be accepted by the client.

Section 2: Banking — Core Concepts & Terminology

Banking clients will be your most common BFSI clients. Understanding how a bank operates will help you contextualize the data you work with.

2.1 Types of Banks

- Retail / Commercial Banks: Serve individuals and businesses. Products include savings accounts, FDs, loans, credit cards.
- Investment Banks: Help companies raise capital, underwrite securities, M&A advisory. Examples: Morgan Stanley, Goldman Sachs.
- Central Banks: Government-owned. Control monetary policy, issue currency, regulate other banks. E.g., RBI (India), Fed (USA), ECB (Europe).
- NBFCs (Non-Banking Financial Companies): Provide banking-like services but don't hold a banking license. E.g., Bajaj Finance, Muthoot Finance.
- Cooperative Banks: Member-owned financial institutions, typically serving rural areas.

2.2 Essential Banking Terminology

Term	Meaning
KYC (Know Your Customer)	The process by which banks verify the identity of their clients. Mandatory for account opening. Involves ID proof, address proof, and PAN/Aadhaar in India.
AML (Anti-Money Laundering)	Processes and systems to detect and prevent money laundering — i.e., disguising illegally obtained money as legitimate income.
CASA Ratio	Current Account and Savings Account ratio. Higher CASA = cheaper funds for the bank, better profitability. Banks love high CASA ratios.
NPA / NPL	Non-Performing Asset / Non-Performing Loan. A loan where the borrower has not paid interest or principal for 90+ days. High NPAs are bad for a bank.
LTV (Loan-to-Value)	Ratio of the loan amount to the appraised value of the asset. E.g., 80% LTV on a home loan means the bank lends 80% of the property value.
EMI (Equated Monthly Installment)	Fixed monthly payment that covers both principal and interest on a loan.
Credit Score / CIBIL Score	A numerical score (300-900 in India) representing a person's creditworthiness. Higher is better. Generated by credit bureaus like CIBIL, Experian, CRIF.
NEFT / RTGS / IMPS / UPI	Payment transfer mechanisms. NEFT: batch-based, RTGS: real-time high-value, IMPS: instant interbank, UPI: instant mobile-based.

Nostro / Vostro Account	Nostro = 'our account held at your bank' (foreign bank). Vostro = 'your account held at our bank'. Used in international banking.
Core Banking System (CBS)	Centralized software platform that processes all daily banking transactions. Examples: Finacle (Infosys), BaNCS (TCS), Temenos.
ALM (Asset Liability Management)	Process of managing the risks arising from mismatches between assets (loans) and liabilities (deposits) in terms of interest rates and maturities.
Net Interest Margin (NIM)	Difference between interest income earned and interest paid to depositors, as a % of assets. Key profitability metric for banks.
Repo Rate	Rate at which the central bank lends money to commercial banks. Increasing repo rate makes borrowing costlier for banks, slowing down lending.
CRR / SLR	Cash Reserve Ratio and Statutory Liquidity Ratio — mandatory reserves banks must maintain with RBI. Regulatory tools to control liquidity.
Churn	When a customer closes their account or stops using a product. Reducing churn is a common data science project in banking.

2.3 Common Data Science Use Cases in Banking

- Credit Scoring & Loan Default Prediction: Build models to predict if a borrower will repay.
- Customer Segmentation: Group customers by behaviour, value, or risk for targeted offers.
- Fraud Detection: Real-time anomaly detection on transactions.
- Churn Prediction: Identify customers likely to leave and trigger retention campaigns.
- Next Best Offer (NBO): Recommend the right product to the right customer at the right time.
- ATM Cash Optimization: Predict cash demand at each ATM to minimize replenishment costs.
- Sentiment Analysis: Analyse customer feedback and social media for NPS improvement.

Section 3: Financial Services — Capital Markets & Beyond

3.1 Capital Markets Overview

Capital markets connect those who need capital (companies, governments) with those who have capital (investors). They are divided into primary and secondary markets.

Term	Meaning
Primary Market	Where new securities are issued for the first time. E.g., IPO (Initial Public Offering). Company raises fresh capital here.
Secondary Market	Where existing securities are bought and sold between investors. E.g., BSE, NSE. The company does not receive money from these trades.
Equity / Stock / Share	Ownership in a company. Shareholders get dividends and capital appreciation. Higher risk, higher potential return.
Bond / Debenture	A debt instrument. The issuer borrows money and promises to pay interest (coupon) and repay principal at maturity. Lower risk than equity.
Mutual Fund	A pool of money from many investors, managed by a professional fund manager, invested in a diversified portfolio of securities.
NAV (Net Asset Value)	Value per unit of a mutual fund. $NAV = \frac{\text{Total Assets} - \text{Liabilities}}{\text{Number of Units}}$
AUM (Assets Under Management)	Total market value of assets that an investment manager manages on behalf of clients. Key size metric for fund houses.
Hedge Fund	A private investment fund using complex strategies (short selling, leverage, derivatives) to generate high returns. Not open to general public.
Derivatives	Financial contracts whose value is derived from an underlying asset. Types: Futures, Options, Swaps, Forwards.
Options (Call/Put)	Call option = right to BUY an asset at a set price. Put option = right to SELL an asset at a set price. Used for hedging and speculation.
Portfolio	A collection of financial assets (stocks, bonds, cash) held by an investor.
Alpha / Beta	Alpha = excess return above benchmark. Beta = sensitivity of asset to market movements. Beta > 1 means more volatile than market.
Volatility	Measure of price fluctuation. High volatility = high risk. Often measured using standard deviation or VIX (Volatility Index).
Liquidity	How easily an asset can be converted to cash without affecting its price. Stocks of large companies are highly liquid.
Bull Market / Bear Market	Bull = rising prices, optimistic sentiment. Bear = falling prices, pessimistic sentiment.
FII / DII	FII = Foreign Institutional Investors (e.g., foreign funds investing in India). DII = Domestic Institutional Investors (e.g., LIC, domestic mutual funds).

3.2 Payments & Fintech Terminology

- POS (Point of Sale): Terminal where card transactions are processed at a merchant.
- Payment Gateway: Technology that authorizes credit/debit card payments in online transactions (e.g., Razorpay, PayU, Stripe).
- Wallet: Digital wallet to store money electronically (e.g., Paytm, Google Pay).
- Interchange Fee: Fee paid by the merchant's bank to the cardholder's bank on every card transaction.
- Acquiring Bank: The bank that processes card payments on behalf of the merchant.
- Issuing Bank: The bank that issued the credit/debit card to the customer.
- BIN (Bank Identification Number): First 6 digits of a card number, identifying the issuing bank.

3.3 Common Data Science Use Cases in Financial Services

- Algorithmic Trading: Build models to execute trades based on signals and market data.
- Portfolio Optimization: Use mean-variance or ML models to construct optimal portfolios.
- Credit Risk Modeling: PD (Probability of Default), LGD (Loss Given Default), EAD (Exposure at Default) — the three pillars of credit risk.
- Market Risk Modeling: VaR (Value at Risk) — estimate maximum potential loss in a portfolio.
- Regulatory Reporting Automation: Automate the generation of Basel/IFRS9 reports.
- Transaction Monitoring: Flag suspicious transactions for AML investigations.

Section 4: Insurance — Core Concepts & Terminology

4.1 How Insurance Works

Insurance is fundamentally about risk pooling. Many people pay small premiums so that those who suffer losses can be compensated. The insurer must accurately price risk to remain profitable.

Term	Meaning
Premium	The amount paid by the policyholder to the insurer, usually monthly or annually, to keep the insurance policy active.
Policy	The legal contract between the insurer and the policyholder outlining coverage, exclusions, and terms.
Policyholder / Insured	The person or entity who owns the insurance policy and pays premiums.

Beneficiary	The person who receives the insurance payout (claim) if the insured event occurs.
Claim	A formal request by the policyholder to the insurance company for payment following a covered loss or event.
Underwriting	The process of evaluating risk and deciding whether to insure it, and at what premium. Underwriters are the risk assessors.
Actuarial Science	The discipline using mathematics and statistics to assess risk in insurance. Actuaries calculate premiums, reserves, and liabilities.
Loss Ratio	Claims paid / Premiums earned. A ratio below 100% means the insurer is profitable from underwriting. Key performance metric.
Combined Ratio	Loss Ratio + Expense Ratio. Below 100% = underwriting profit. Above 100% = underwriting loss (may still profit from investments).
Deductible / Excess	Amount the policyholder must pay out-of-pocket before insurance kicks in. Higher deductible = lower premium.
Sum Assured / Coverage	The maximum amount the insurer will pay in case of a covered event.
Lapse / Surrender	Lapse = policy terminates due to non-payment of premium. Surrender = policyholder voluntarily cancels the policy early.
Reinsurance	Insurance for insurance companies. Insurers transfer a portion of their risk to reinsurers to reduce their exposure.
Mortality Rate	Used in life insurance — the rate of death within a specific population over a defined period. Basis for life insurance pricing.
No Claim Bonus (NCB)	Discount offered on premium renewal if no claims were made in the previous policy year.
Third Party Liability	Coverage for damage caused to another person or their property. Mandatory in vehicle insurance in India.
IRDA / IRDAI	Insurance Regulatory and Development Authority of India. Regulates the insurance sector in India.

4.2 Types of Insurance Products

- Life Insurance: Term life, Whole life, Endowment, ULIPs (Unit Linked Insurance Plans).
- Health Insurance: Individual, Family floater, Group health (employer-provided), Critical illness.
- General Insurance: Motor, Home, Travel, Marine, Fire insurance.
- Crop Insurance: Coverage for farmers against crop failure — important in India (PMFBY scheme).

4.3 Common Data Science Use Cases in Insurance

- Fraud Detection: Identify fraudulent claims using anomaly detection and ML classifiers.
- Claims Prediction: Predict claim frequency and severity for pricing and reserving.
- Telematics / Usage-Based Insurance: Use driving behavior data from IoT devices to price motor insurance dynamically.
- Customer Lifetime Value (CLV): Predict long-term value of a policyholder to optimize acquisition spend.
- Churn / Lapse Prediction: Identify policyholders likely to lapse and trigger retention actions.
- Underwriting Automation: Use ML to automate risk scoring for policy applications.

Section 5: Regulatory Framework & Compliance

Regulation is central to BFSI. As a data scientist, your models and outputs must be compliant. Clients will often ask about regulatory requirements.

5.1 Key Regulatory Bodies

Country / Region	Regulator	Jurisdiction
India	RBI (Reserve Bank of India)	Banks, NBFCs, payment systems
India	SEBI (Securities and Exchange Board of India)	Capital markets, mutual funds, brokers
India	IRDAI	Insurance companies
India	PFRDA	Pension funds
USA	Federal Reserve / OCC / FDIC	Banks and financial institutions
USA	SEC (Securities & Exchange Commission)	Capital markets
Global	FATF (Financial Action Task Force)	AML and counter-terrorism financing standards

5.2 Key Regulatory Frameworks

Basel III / Basel IV
International banking regulations setting standards for capital adequacy, stress testing, and liquidity. Banks must hold sufficient capital (CET1, Tier 1, Tier 2) relative to their risk-

weighted assets. Key metrics: CAR (Capital Adequacy Ratio), LCR (Liquidity Coverage Ratio), NSFR (Net Stable Funding Ratio).

IFRS 9 (International Financial Reporting Standard 9)

Accounting standard requiring banks to provision for expected credit losses (ECL) proactively, in three stages: Stage 1 (performing loans), Stage 2 (underperforming), Stage 3 (NPLs). Replaced the older incurred-loss model. Significant driver of credit risk modeling projects.

GDPR / Data Privacy Regulations

GDPR (Europe) and equivalent laws require firms to protect customer data, obtain consent, and allow data deletion. In India, the DPDP Act (Digital Personal Data Protection Act) governs this. As a data scientist, always anonymize / pseudonymize PII (Personally Identifiable Information) in your datasets.

AML / KYC / CFT Compliance

Banks must implement programs to detect and report suspicious activity (SARs — Suspicious Activity Reports). Transaction monitoring systems flag unusual patterns. Customer Risk Rating (CRR) classifies customers by money laundering risk. CFT = Combating the Financing of Terrorism.

Section 6: Key Metrics & KPIs You Will Encounter

When presenting data science results to BFSI clients, you need to speak in their language. Here are the most common business metrics they care about.

6.1 Banking KPIs

KPI	Definition
NIM (Net Interest Margin)	$(\text{Interest Income} - \text{Interest Expense}) / \text{Avg Earning Assets}$. Profitability from lending.
NPL Ratio / GNPA Ratio	Gross NPA / Total Loans. Lower is better. Key asset quality metric.
PCR (Provision Coverage Ratio)	Provisions held / Gross NPAs. Shows how much bad debt is covered by provisions.

ROA (Return on Assets)	Net Profit / Total Assets. Measures how efficiently assets are used to generate profit.
ROE (Return on Equity)	Net Profit / Shareholders' Equity. Returns generated for shareholders.
CAR (Capital Adequacy Ratio)	Capital / Risk-Weighted Assets. Minimum 10.5% required under Basel III.
CASA Ratio	CASA Deposits / Total Deposits. Higher = cheaper funding for the bank.
Cost-to-Income Ratio	Operating Expenses / Operating Income. Lower is better. Efficiency metric.

6.2 Credit Risk Metrics

Metric	Definition
PD (Probability of Default)	Likelihood that a borrower will default within a given time horizon. Core output of credit scorecards.
LGD (Loss Given Default)	Percentage of the exposure the bank loses if the borrower defaults. E.g., 40% LGD means recovery of 60%.
EAD (Exposure at Default)	Total value exposed to loss at time of default. For revolving credit (like credit cards), EAD can be higher than current balance.
Expected Loss (EL)	$EL = PD \times LGD \times EAD$. The average credit loss a bank expects to incur.
Vintage Analysis	Tracking loan cohorts by their month/year of origination to compare default patterns over time.
Roll Rate / Roll Rate Analysis	Percentage of accounts moving from one delinquency bucket to the next (e.g., 30DPD to 60DPD). Used in collections strategy.
DPD (Days Past Due)	Number of days a payment is overdue. Buckets: 0 DPD (current), 30 DPD, 60 DPD, 90+ DPD (NPA).

6.3 Insurance KPIs

KPI	Definition
Loss Ratio	Net Claims / Net Premiums Earned. Target varies by line; typically < 70-75%.
Combined Ratio	Loss Ratio + Expense Ratio. < 100% = underwriting profit.

GWP (Gross Written Premium)	Total premiums written before deducting reinsurance. Top-line revenue metric.
Renewal Retention Rate	% of policies that renew at expiry. High retention = lower acquisition costs.
Claims Frequency	Number of claims per 100 policies in a year.
Claims Severity	Average cost per claim.
Solvency Ratio	Available Solvency Margin / Required Solvency Margin. Must be > 150% per IRDAI regulations.

Section 7: Data Landscape in BFSI

7.1 Types of Data You Will Work With

- Transaction Data: Every credit, debit, transfer — highly voluminous, real-time or near-real-time.
- Customer Master Data: Demographics, KYC details, account information, contact data.
- Credit Bureau Data: CIBIL/Experian scores, credit history, enquiries, existing loans.
- Market Data: Stock prices, bond yields, FX rates, commodity prices — time-series data.
- Claims Data (Insurance): Claim date, type, amount, settlement status, fraud flags.
- Policy Data (Insurance): Policy terms, premium, coverage, rider details.
- Behavioral Data: Login patterns, channel usage, browsing behavior (digital banking).
- Alternative Data: Social media, geolocation, utility payment history — for thin-file customers.

7.2 Common Systems & Technologies

System / Term	What It Is
Core Banking System (CBS)	The central transaction processing platform (Finacle, BaNCS, T24/Temenos, Oracle FLEXCUBE).
CRM (Customer Relationship Management)	System to manage customer interactions and data. E.g., Salesforce, Microsoft Dynamics.
Data Warehouse / EDW	Central repository of integrated data from multiple sources for analytics and reporting. E.g., Teradata, Snowflake, Redshift.
Data Lake	Storage repository that holds vast amounts of raw data in native format until needed. E.g., Azure Data Lake, S3-based on AWS.

LOS (Loan Origination System)	System that manages the entire loan application process from application to disbursement.
LMS (Loan Management System)	System for managing active loans — EMI tracking, prepayments, NPA classification.
Policy Administration System (PAS)	Core system in insurance for managing policies throughout their lifecycle.
Claims Management System	Tracks insurance claims from First Notice of Loss (FNOL) to settlement.
FTP (Funds Transfer Pricing)	Internal pricing mechanism to allocate interest income across business units based on the cost of funds.
SWIFT	Global messaging network for international financial transactions. SWIFT codes identify banks internationally.

Section 8: Quick Reference Glossary — A to Z

A rapid-fire glossary of terms you may encounter in client conversations or documents.

Term / Abbreviation	Quick Definition
ALCO	Asset-Liability Committee — senior committee managing balance sheet risk.
AMC	Asset Management Company — manages mutual funds. E.g., HDFC AMC, SBI Mutual Fund.
ATM (in banking context)	Automated Teller Machine. Also used: White-label ATM (non-bank owned), Brown-label ATM.
BIS	Bank for International Settlements — coordinates global banking regulation standards.
CAGR	Compound Annual Growth Rate — year-over-year growth rate over a period.
CET1	Common Equity Tier 1 — highest quality capital (ordinary shares, retained earnings) per Basel III.
CLO / CDO / MBS	Collateralized Loan Obligation / Collateralized Debt Obligation / Mortgage-Backed Security — structured financial products.
COGS (for NBFCs)	Cost of Goods Sold — used loosely to mean cost of funds in financial services context.
Collateral	Asset pledged as security against a loan. If borrower defaults, lender can seize collateral.

Credit Card Cycle	Billing period (usually monthly) — purchase date, statement date, due date, payment date.
DSCR	Debt Service Coverage Ratio = Net Operating Income / Total Debt Service. Ability to repay debt.
EBIDTA (loosely)	Earnings used to assess financial health of a corporate client in credit analysis.
FATCA	Foreign Account Tax Compliance Act — US law requiring foreign banks to report US clients' accounts to the IRS.
FEMA	Foreign Exchange Management Act — governs foreign exchange in India (RBI oversees).
FIFO / LIFO	Accounting methods for valuing inventory / loan repayments.
GIC	Guaranteed Investment Contract — insurance product guaranteeing rate of return.
GNPA / NNPA	Gross NPA / Net NPA (after deducting provisions). Key asset quality ratios.
HNI / UHNI	High Net-worth Individual (>1 Cr investable) / Ultra High Net-worth Individual (>5 Cr). Premium banking segment.
IRDA Circular	Regulatory direction from IRDAI that insurers must comply with.
Lien	Legal right of a lender to retain possession of an asset as security until a debt is repaid.
MCLR	Marginal Cost of Funds Based Lending Rate — benchmark rate for bank loans in India.
Moratorium	Temporary suspension of loan repayment obligations. E.g., COVID-19 moratorium (2020).
NACH	National Automated Clearing House — used for bulk recurring payment transactions (like EMI auto-debit) in India.
OCR	Optical Character Recognition — used to digitize physical KYC documents, loan applications.
P&L (in banking)	Profit and Loss statement. Also: Provisions and Losses in a credit context.
PAR (Portfolio at Risk)	% of loan portfolio with payments overdue > 30 days. Microfinance / NBFC metric.
Prudential Norms	RBI's regulations on income recognition, asset classification, and provisioning for banks.
ROI (in lending)	Rate of Interest charged on a loan. Different from the general business meaning.
SAR	Suspicious Activity Report — filed by financial institutions when they detect potential money laundering.

Securitization	Pooling assets (like home loans) and selling them as securities to investors to raise funds.
Stress Testing	Simulating extreme scenarios (economic crash, credit crisis) to test portfolio resilience. Regulatory requirement.
Tenor / Tenure	Duration of a loan or investment.
TDS	Tax Deducted at Source — tax deducted by the bank on interest income before crediting to customer.
UW Score	Underwriting Score — composite score used by insurance underwriters to assess risk.
Working Capital	Short-term funds needed by a business to meet day-to-day expenses. Banks offer working capital loans / OD facilities.
Write-off	When a bank removes an NPA from its books after exhausting recovery efforts. The debt still legally exists.
XBRL	eXtensible Business Reporting Language — standard format for submitting financial reports to regulators.
Yield	Income generated by an investment, expressed as % of cost or current value. Bond yield = annual coupon / price.

Section 9: Tips for Working with BFSI Clients

Do's for Data Scientists in BFSI

- Always ask the client to clarify acronyms when unsure — it's better to ask than to assume.
- Understand the business problem first, then think about the ML model. Ask: 'What decision will this model support?'
- Be cautious with PII data — never use raw customer data without proper anonymization.
- Present model results in business terms: 'This model will reduce NPA by X% in the first year' beats 'AUC = 0.87'.
- Model interpretability matters — regulators may require explanations for credit decisions (Right to Explanation).
- Understand the data lineage — where does this data come from? CBS, CRM, bureau? Know your data sources.

Watch Out For

- Data Quality Issues: BFSI data can have gaps, inconsistencies, and legacy system artifacts. Always validate.
- Temporal Leakage: In credit models, never use future data (e.g., 12-month-forward delinquency) as a feature.
- Class Imbalance: Fraud data and default data are typically highly imbalanced. Use SMOTE, undersampling, or adjusted thresholds.
- Model Fairness / Bias: Credit models must not discriminate based on protected attributes (gender, religion, geography).
- Regulatory Timelines: Clients often have hard regulatory submission deadlines. Always clarify timelines early.

You are now equipped with the foundational knowledge to confidently navigate BFSI client engagements. Good luck!
